

PEACEABLE QUEENS ON EVEN TORI: AN EXACT PARITY FORMULA

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ABSTRACT. Let $t(n)$ be the largest m for which m black and m white queens can be placed on the $n \times n$ toroidal chessboard with no black–white attack. We prove that, for every positive integer q ,

$$t(2q) = \max_{(r_0, r_1, c_0, c_1) \in \{0, \dots, q\}^4} \min\{r_0 c_1 + r_1 c_0, (q - r_0)(q - c_0) + (q - r_1)(q - c_1)\}.$$

The maximizing construction separates the two square parities and extends the plaid idea of Clinch, Drescher, Huynh and Saffidine. The upper bound combines an exact moment relaxation, rational branch certificates, a hand-checkable algebraic finish for $q \geq 130$, and a rerunnable exact integer cut-envelope computation for the finite range. In particular,

$$\lim_{\substack{n \rightarrow \infty \\ 2|n}} \frac{t(n)}{n^2} = \frac{2 - \sqrt{3}}{2} = 0.1339745\dots$$

We also prove the odd values $t(15) = 20$ and $t(17) = 28$ by exhaustive finite searches with separately written audits.

1. INTRODUCTION

A peaceable placement consists of equally many black and white queens, with no queen attacking a queen of the other colour; attacks within an army are allowed. The problem goes back to Ainley [1]. On the ordinary board its asymptotic density remains open [3]; on the toroidal board the rows, columns and diagonals wrap around, and the problem becomes translation invariant. Its maxima form OEIS sequence A279405 [4].

Clinch, Drescher, Huynh and Saffidine [3, 2] introduced *plaid* constructions on even tori. They obtained asymptotic lower density 0.1339 and upper density 0.1402, while for odd orders they proved the upper bound $t(n) \leq n^2/8$. The following theorem determines the even problem exactly.

Definition 1. For a positive integer q , put

$$(1) \quad H(2q) := \max_{(r_0, r_1, c_0, c_1) \in \{0, \dots, q\}^4} \min\{r_0 c_1 + r_1 c_0, (q - r_0)(q - c_0) + (q - r_1)(q - c_1)\}.$$

Theorem 2. For every positive integer q ,

$$t(2q) = H(2q).$$

Corollary 3. Let $\kappa = 4 - 2\sqrt{3}$ and $\gamma = 1 - 1/\sqrt{3}$. For every positive integer q ,

$$\kappa q^2 - \gamma q - \frac{1}{4} \leq H(2q) \leq \kappa q^2.$$

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Consequently, for even n ,

$$t(n) = \frac{2-\sqrt{3}}{2} n^2 + O(n), \quad \lim_{\substack{n \rightarrow \infty \\ 2|n}} \frac{t(n)}{n^2} = \frac{2-\sqrt{3}}{2} = 0.1339745\dots,$$

and $t(n) \leq \lfloor (2-\sqrt{3})n^2/2 \rfloor$.

Corollary 4. *Every even value of A279405 is computable from (1); for example,*

$$t(16) = 32, \quad t(18) = 42, \quad t(20) = 51, \quad t(40) = 210.$$

Theorem 5. $t(15) = 20$ and $t(17) = 28$.

Theorem 2 says that an optimum always has a *parity-separated* representative: the diagonal colours reserve one square parity for each army, leaving the four row and column parity counts in (1). This family contains the classical plaids but is strictly larger; for example it gives $H(20) = 51$, whereas direct optimization of the classical two-parameter subfamily gives 50.

For the upper bound we first replace queens by a colouring of the four line families. On an even torus, conditioning on row and column parities yields a 42-equation moment relaxation. Two exact rational branch certificates localize every possible counterexample to a small neighbourhood of the parity-separated profile. Two support inequalities then reduce that neighbourhood to a short algebraic argument for $q \geq 130$. The remaining orders are covered by a rerunnable exact integer cut-envelope computation, apart from five small orders settled by direct exhaustive search. Section 7 gives a proof-complete account of the searches for Theorem 5. Appendix A describes the released certificates, audits and reproduction commands. All proof decisions use integer or rational arithmetic.

2. THE FORMULA AND THE LOWER BOUND

2.1. Line colourings. Throughout, $n \geq 2$, all coordinates lie in $\mathbb{Z}/n\mathbb{Z} = \{0, 1, \dots, n-1\}$, and the board is $G = (\mathbb{Z}/n\mathbb{Z})^2$. Through a cell (r, c) pass four *lines*: row r , column c , diagonal $d = r - c$ and antidiagonal $a = r + c$, each family indexed by $\mathbb{Z}/n\mathbb{Z}$. Two queens attack each other exactly when they share a line. For $R, C, D, A \subseteq \mathbb{Z}/n\mathbb{Z}$ put

$$B(R, C, D, A) = \{(r, c) \in G : r \in R, c \in C, r - c \in D, r + c \in A\},$$

and $W(R, C, D, A) = B(R^c, C^c, D^c, A^c)$, complements taken family by family.

Lemma 6 (Line-colouring equivalence). *Let $m \geq 1$. A peaceable placement of m black and m white queens on the $n \times n$ torus exists if and only if there are $R, C, D, A \subseteq \mathbb{Z}/n\mathbb{Z}$ with $|B| \geq m$ and $|W| \geq m$.*

Proof. ($\Rightarrow$) Colour every line carrying a black queen black and every other line white; no line carries queens of both colours, else those two queens attack. Every black queen then lies on four black lines and every white queen on four white lines, and the queens of one colour occupy distinct cells.

($\Leftarrow$) Place black queens on any m cells of B and white queens on any m cells of W . The two sets are disjoint, and a black–white attack would need a line lying simultaneously in a black set and in its complement. $\square$

Consequently $t(n) = \max_{R, C, D, A} \min(|B|, |W|)$, and the queens have disappeared. Complementing all four sets exchanges B and W , so $\min(|B|, |W|)$ is invariant under complementation; we use this freely.

Lemma 7 (Incidence). *Let ℓ, ℓ' be lines of two distinct families. Unless they form a diagonal-antidiagonal pair, they meet in exactly one cell. A diagonal d and an antidiagonal a meet in the fibre*

$$\varphi^{-1}(d, a), \quad \varphi(r, c) = (r - c, r + c).$$

Every point in this fibre satisfies $2r = a + d$ and $2c = a - d$; these doubled congruences are equivalent to the original pair only when n is odd. The fibre has one cell for odd n . For even n it has two cells, differing by $(n/2, n/2)$, when $d \equiv a \pmod{2}$, and is empty otherwise.

Proof. The first statement is immediate (a row r and a column c meet in (r, c) ; a row r and a diagonal d in $(r, r - d)$; and so on). For the second, consider $\varphi(r, c) = (r - c, r + c)$ on G . Its kernel is $\{(r, c) : r = c, 2r = 0\}$, trivial for odd n and of order 2 for even n , generated by $(\frac{n}{2}, \frac{n}{2})$. For even n every nonempty fibre thus has two elements and the image has $n^2/2$ elements; every image point satisfies $d \equiv a \pmod{2}$, and there are exactly $n^2/2$ such pairs, so the image is precisely that set. $\square$

Lemma 8 (Budget). *If $|B| \geq m$ and $|W| \geq m$ for some (R, C, D, A) , then the same holds for a configuration satisfying*

$$|R| + |C| + |D| + |A| \leq 2n.$$

Proof. Complementing all four sets exchanges B and W and replaces the sum of their sizes by $4n$ minus that sum. $\square$

For two line families X, Y , let $I(X, Y)$ denote the number of cells whose lines in both families are black.

Lemma 9 (Triple identity). *Choose three line families with black sets of sizes x, y, z . If U and U' count the cells whose three lines are all black and all white, respectively, then*

$$(2) \quad U + U' = n^2 - n(x + y + z) + I(X, Y) + I(X, Z) + I(Y, Z).$$

In particular, $|B|, |W| \geq m$ implies that the right-hand side is at least $2m$.

Proof. For the three black-line indicators u, v, w on the cells,

$$uvw + (1 - u)(1 - v)(1 - w) = 1 - u - v - w + uv + uw + vw.$$

Summing gives (2); moreover, $B \subseteq U$ and $W \subseteq U'$. $\square$

2.2. The parity-separated construction. Now let $n = 2q$. The diagonal and antidiagonal labels through (r, c) have the same parity, namely that of $r + c$. Colouring both diagonal families black on one label parity and white on the other therefore reserves one square parity for each army. Only the row and column choices remain.

Split each of the four families by label parity: for $i \in \{0, 1\}$ let $r_i = |\{\rho \in R : \rho \equiv i\}|$, and likewise c_i, d_i, a_i ; each parity class of $\mathbb{Z}/2q\mathbb{Z}$ has exactly q labels, so all eight counts lie in $\{0, \dots, q\}$.

Proposition 10 (Parity separation). *Let $q \geq 1$ and $0 \leq r_0, r_1, c_0, c_1 \leq q$. Choose $R \subseteq \mathbb{Z}/2q\mathbb{Z}$ consisting of any r_0 even and any r_1 odd labels, C of any c_0 even and c_1 odd labels, and set $D = A = \{\text{odd labels}\}$. Then*

$$|B| = r_0 c_1 + r_1 c_0, \quad |W| = (q - r_0)(q - c_0) + (q - r_1)(q - c_1).$$

Consequently $t(2q) \geq H(2q)$.

Proof. A cell (r, c) is black-homogeneous iff $r \in R$, $c \in C$ and both $r - c$ and $r + c$ are odd, i.e. iff $r \in R$, $c \in C$ and r, c have opposite parities; there are $r_0c_1 + r_1c_0$ such cells. It is white-homogeneous iff $r \notin R$, $c \notin C$ and $r + c$ is even, i.e. r, c of equal parity; there are $(q - r_0)(q - c_0) + (q - r_1)(q - c_1)$ such cells. Now apply Lemma 6 with $m = \min(|B|, |W|)$ and maximize over the four counts. $\square$

Remark 11. The construction extends the plaid mechanism of [2, §3]. Up to exchanging parity classes, the classical two-parameter subfamily has profiles (u, q, v, q) and hence capacity

$$\min\{q(u + v), (q - u)(q - v)\}, \quad 0 \leq u, v \leq q.$$

The four counts in Proposition 10 are independent, so its family is larger. Maximizing the displayed two-variable expression gives 50 at $n = 20$ and 72 at $n = 24$, whereas $H(20) = 51$ and $H(24) = 74$. At $n = 16$, the profile $(0, 6, 6, 0)$ gives 36 black-homogeneous and 32 white-homogeneous cells.

n	2	4	6	8	10	12	14	16	18	20
$H(n)$	0	2	4	8	12	18	25	32	42	51
n	22	24	26	28	30	32	34	36	38	40
$H(n)$	64	74	90	101	120	133	153	170	190	210

TABLE 1. Exact even values through order 40. The entries through order 14 agree with A279405, accessed July 27, 2026.

2.3. Values and size of H .

Lemma 12 (Continuous upper bound). *Let $q > 0$ be real and let $r_0, r_1, c_0, c_1 \in [0, q]$ be real. Put*

$$X = r_0c_1 + r_1c_0, \quad Y = (q - r_0)(q - c_0) + (q - r_1)(q - c_1).$$

Then $\min(X, Y) \leq \kappa q^2$, where $\kappa = 4 - 2\sqrt{3}$. In particular $H(2q) \leq \lfloor \kappa q^2 \rfloor = \lfloor \frac{2-\sqrt{3}}{2}(2q)^2 \rfloor$.

Proof. Normalize: $\tilde{r}_i = r_i/q$, $\tilde{c}_i = c_i/q$, and set

$$\rho = \tilde{r}_0 + \tilde{r}_1, \quad \chi = \tilde{c}_0 + \tilde{c}_1, \quad \xi = 1 - \rho, \quad \eta = 1 - \chi, \quad z = (\tilde{r}_0 - \tilde{r}_1)(\tilde{c}_0 - \tilde{c}_1).$$

Expanding,

$$(3) \quad \frac{X - Y}{q^2} = \rho + \chi - 2 - z, \quad \frac{X + Y}{q^2} = 2 - \rho - \chi + \rho\chi = 1 + \xi\eta,$$

and $|z| \leq \rho\chi$ because $|\tilde{r}_0 - \tilde{r}_1| \leq \tilde{r}_0 + \tilde{r}_1 = \rho$ and likewise for the columns.

If $\xi\eta \leq 0$ then $\min(X, Y) \leq \frac{X+Y}{2} \leq \frac{q^2}{2} < \kappa q^2$.

The map $(r_0, r_1, c_0, c_1) \mapsto (q - r_0, q - r_1, q - c_1, q - c_0)$ preserves the domain, exchanges X and Y , and replaces (ξ, η) by $(-\xi, -\eta)$. So we may assume $\xi, \eta \geq 0$.

Suppose first $X \geq Y$. Then $z \leq \rho + \chi - 2$ by (3), and $z \geq -\rho\chi$, so $2 - \rho - \chi \leq \rho\chi$, i.e. $\xi + \eta \leq 1 - \xi - \eta + \xi\eta$, i.e.

$$(4) \quad 2\xi + 2\eta - \xi\eta \leq 1.$$

Put $w = \sqrt{\xi\eta} \in [0, 1]$. By AM-GM, $4w - w^2 \leq 2\xi + 2\eta - \xi\eta \leq 1$, so $w^2 - 4w + 1 \geq 0$ and hence $w \leq 2 - \sqrt{3}$. By (3),

$$\min(X, Y) \leq \frac{X + Y}{2} = q^2 \frac{1 + w^2}{2} \leq q^2 \frac{1 + (2 - \sqrt{3})^2}{2} = (4 - 2\sqrt{3})q^2.$$

Now suppose $Y \geq X$, so $\min(X, Y) = X$, and assume for contradiction $X > \kappa q^2$. Then $X + Y \geq 2X > 2\kappa q^2$, so by (3) $\xi\eta > 2\kappa - 1 = (2 - \sqrt{3})^2$ and $w > 2 - \sqrt{3}$. On the other hand $X/q^2 = \tilde{r}_0\tilde{c}_1 + \tilde{r}_1\tilde{c}_0$ is bilinear in $(\tilde{r}_0, \tilde{r}_1)$ and $(\tilde{c}_0, \tilde{c}_1)$ subject to the fixed sums ρ, χ , hence is at most $\rho\chi$ (attained at a vertex), and $\rho\chi = (1 - \xi)(1 - \eta) \leq (1 - w)^2$ by AM–GM. Therefore

$$\kappa < \frac{X}{q^2} \leq (1 - w)^2 < (1 - (2 - \sqrt{3}))^2 = (\sqrt{3} - 1)^2 = 4 - 2\sqrt{3} = \kappa,$$

a contradiction. Finally, $H(2q)$ is an integer maximum of $\min(X, Y)$ over a subset of the domain, so $H(2q) \leq \lfloor \kappa q^2 \rfloor$. $\square$

Lemma 13. *Let $s = \sqrt{3} - 1$, so $\kappa = s^2 = 4 - 2\sqrt{3}$, and let $\gamma = 1 - 1/\sqrt{3} = s/(1 + s)$. Then for every $q \geq 1$,*

$$H(2q) \geq \kappa q^2 - \gamma q - \frac{1}{4}.$$

Proof. For an integer $0 \leq P \leq 2q$ take the profile $(r_0, r_1, c_0, c_1) = (0, \lfloor P/2 \rfloor, \lceil P/2 \rceil, 0)$. Then

$$X = \left\lfloor \frac{P}{2} \right\rfloor \left\lceil \frac{P}{2} \right\rceil = \left\lfloor \frac{P^2}{4} \right\rfloor, \quad Y = q \left(q - \left\lceil \frac{P}{2} \right\rceil \right) + \left(q - \left\lfloor \frac{P}{2} \right\rfloor \right) q = 2q^2 - qP,$$

so $H(2q) \geq \min(\lfloor P^2/4 \rfloor, 2q^2 - qP)$. The two branches cross where $P^2/4 = 2q^2 - qP$, i.e. at $P = 2sq$, with common value $s^2 q^2 = \kappa q^2$.

Write $2sq = m + \theta$ with $m = \lfloor 2sq \rfloor$ and $\theta \in [0, 1)$. Taking $P = m \leq 2sq$ gives $X \geq P^2/4 - \frac{1}{4} = \kappa q^2 - sq\theta + \theta^2/4 - \frac{1}{4} \geq \kappa q^2 - sq\theta - \frac{1}{4}$ and $Y = 2q^2 - qm = \kappa q^2 + q\theta \geq \kappa q^2$, so the value is at least $\kappa q^2 - sq\theta - \frac{1}{4}$. Taking $P = m + 1 \geq 2sq$ gives $X \geq P^2/4 - \frac{1}{4} \geq \kappa q^2 - \frac{1}{4}$ and $Y = \kappa q^2 - q(1 - \theta)$, so the value is at least $\kappa q^2 - q(1 - \theta) - \frac{1}{4}$. Hence

$$H(2q) \geq \kappa q^2 - q \min(s\theta, 1 - \theta) - \frac{1}{4} \geq \kappa q^2 - q \max_{0 \leq \theta < 1} \min(s\theta, 1 - \theta) - \frac{1}{4},$$

and the inner maximum is attained at $s\theta = 1 - \theta$, with value $s/(1 + s) = 1 - 1/\sqrt{3} = \gamma$. $\square$

Lemmas 12 and 13 give Corollary 3 once Theorem 2 is proved. Numerically $\gamma = 0.42264\dots$, so $H(2q)$ deviates from κq^2 by less than $\frac{1}{2}q + \frac{1}{4}$.

3. THE UPPER BOUND: RELAXATION AND CANONICAL CHAMBER

From here to the end of Section 5, $n = 2q$ and we prove $t(2q) \leq H(2q)$ for $q \geq 130$. Fix a configuration (R, C, D, A) with black count $B = |B|$ and white count $W = |W|$, and let

$$(r_0, r_1, c_0, c_1, d_0, d_1, a_0, a_1) \in \{0, \dots, q\}^8$$

be its eight parity-split outer counts, with normalization $x = (r_0, \dots, a_1)/q \in [0, 1]^8$.

3.1. The 42-equation moment relaxation. Partition the board into four *row/column parity blocks* $G_{ij} = \{(r, c) : r \equiv i, c \equiv j \pmod{2}\}$, each of q^2 cells. Inside G_{ij} every cell has square parity $e = i \oplus j$, so its diagonal and antidiagonal labels both have parity e ; and each of the four line families restricted to the relevant parity class has exactly q labels.

For each block introduce 16 nonnegative variables

$$p_{ij}(\varepsilon_R, \varepsilon_C, \varepsilon_D, \varepsilon_A), \quad (\varepsilon_R, \varepsilon_C, \varepsilon_D, \varepsilon_A) \in \{0, 1\}^4,$$

intended as q^{-2} times the number of cells of G_{ij} whose row, column, diagonal and antidiagonal are black (1) or white (0) as indicated. Write $E_{ij}[\cdot]$ for the corresponding expectation. A genuine configuration satisfies the following 42 equations, in which $e = i \oplus j$.

Lemma 14 (Soundness of the moment system). *For every configuration and every block (i, j) ,*

$$(5) \quad \sum p_{ij} = 1; \quad E_{ij}[\varepsilon_R] = \frac{r_i}{q}, \quad E_{ij}[\varepsilon_C] = \frac{c_j}{q}, \quad E_{ij}[\varepsilon_D] = \frac{d_e}{q}, \quad E_{ij}[\varepsilon_A] = \frac{a_e}{q};$$

$$(6) \quad E_{ij}[\varepsilon_R \varepsilon_C] = \frac{r_i c_j}{q^2}, \quad E_{ij}[\varepsilon_R \varepsilon_D] = \frac{r_i d_e}{q^2}, \quad E_{ij}[\varepsilon_R \varepsilon_A] = \frac{r_i a_e}{q^2},$$

$$E_{ij}[\varepsilon_C \varepsilon_D] = \frac{c_j d_e}{q^2}, \quad E_{ij}[\varepsilon_C \varepsilon_A] = \frac{c_j a_e}{q^2};$$

and, for each square parity $e \in \{0, 1\}$, the aggregate equation

$$(7) \quad \sum_{i \oplus j = e} E_{ij}[\varepsilon_D \varepsilon_A] = \frac{2 d_e a_e}{q^2}.$$

These are $4 \cdot 10 + 2 = 42$ equations, whose right-hand sides involve exactly 22 distinct products of outer counts. Moreover

$$(8) \quad \frac{B}{q^2} = \sum_{i,j} p_{ij}(1, 1, 1, 1), \quad \frac{W}{q^2} = \sum_{i,j} p_{ij}(0, 0, 0, 0).$$

Proof. The normalization and the four marginals in (5) hold because within G_{ij} each of the four families' relevant parity classes is swept uniformly: for instance the map $G_{ij} \rightarrow (\mathbb{Z}_q)$, $(r, c) \mapsto (\text{index of } r \text{ within its parity class})$, has all fibres of size q .

For (6) we need each pair of families other than $\{D, A\}$ to fibre uniformly over G_{ij} . Writing $r = 2r' + i$, $c = 2c' + j$ with $r', c' \in \mathbb{Z}_q$, the pair (row, column) corresponds to (r', c') , the pair (row, diagonal) to $(r', r' - c')$, (row, antidiagonal) to $(r', r' + c')$, (column, diagonal) to $(c', r' - c')$ and (column, antidiagonal) to $(c', r' + c')$. Each of these is an invertible linear map $\mathbb{Z}_q^2 \rightarrow \mathbb{Z}_q^2$ of determinant ± 1 , hence a bijection; so all q^2 label pairs occur exactly once and the corresponding moment factorizes into the product of the two black-line densities.

The pair $\{D, A\}$ does not fibre uniformly within a block: $(r', c') \mapsto (r' - c', r' + c')$ has determinant 2. But by Lemma 7 each pair (d, a) of labels of equal parity e meets in exactly two cells, and these two cells (r, c) , $(r + q, c + q)$ lie in the two blocks with $i \oplus j = e$ (in the same block iff q is even, in different blocks otherwise — in either case in the union). Summing over the two blocks therefore counts each equal-parity label pair exactly twice, giving (7). Finally (8) is the definition. $\square$

Remark 15. The failure of the $\{D, A\}$ pair to factorize *per block* is not cosmetic: for $4 \mid n$ the two lifts of an equal-parity (d, a) pair land in the *same* block, so per-block independence of the diagonal families is false and 2-adic information survives parity conditioning. Replacing (7) by four per-block equations would be unsound. This is the reason the relaxation has 42 and not 44 rows.

Definition 16. For $x \in [0, 1]^8$ let $\text{OPT}_{42}(x)$ be the maximum of t subject to: $p \geq 0$; the 42 equations of Lemma 14 with the right-hand sides evaluated at x (that is, r_i/q replaced by x -coordinates and each product $r_i c_j / q^2$ by the corresponding product of x -coordinates); and $t \leq \sum_{i,j} p_{ij}(1, 1, 1, 1)$, $t \leq \sum_{i,j} p_{ij}(0, 0, 0, 0)$.

The maximum is well defined. For each block and each x , four independent Bernoulli indicators with the prescribed marginals satisfy all block moments; summing the two relevant blocks also gives each aggregate $\{D, A\}$ equation. Thus the p -feasible set is nonempty. It is compact, and after eliminating t the objective is the minimum of two continuous linear functions of p , so the maximum is attained.

Lemma 14 immediately gives the key inequality

$$(9) \quad \frac{\min(B, W)}{q^2} \leq \text{OPT}_{42}(x)$$

for every configuration with normalized outer profile x .

3.2. The canonical chamber.

Lemma 17 (Chamber). *Every configuration can be transformed, without changing $\{B, W\}$ or the multiset of homogeneous cell counts, into one whose normalized profile x satisfies*

$$(10) \quad r_0 \leq r_1, \quad c_0 \leq c_1, \quad r_0 + r_1 \leq c_0 + c_1, \quad d_0 + d_1 \leq a_0 + a_1, \quad \sum_{i=1}^8 x_i \leq 4.$$

Proof. Complementing all four line sets exchanges B and W and replaces $\sum x_i$ by $8 - \sum x_i$, so the budget can be enforced. Translating the board by one unit in the row direction, $(r, c) \mapsto (r+1, c)$, exchanges the two row parity classes (and simultaneously the two diagonal and the two antidiagonal classes); translating by one unit in the column direction does the same for the columns. Composing the two if necessary, we may sort $r_0 \leq r_1$ and $c_0 \leq c_1$. Transposing, $(r, c) \mapsto (c, r)$, exchanges rows with columns and fixes the antidiagonals while negating the diagonals, hence orders the row and column totals. Finally $(r, c) \mapsto (r, -c)$ fixes the row parities, permutes the column labels within their parity classes, and exchanges the diagonal with the antidiagonal family, so it orders $d_0 + d_1 \leq a_0 + a_1$. Each of these is an affine bijection of the board mapping lines to lines, so B and W are preserved (or exchanged, in the complementation step). $\square$

The four ordering inequalities in (10), apart from the budget row, preserve the sorting achieved by the earlier steps. Indeed, a unit translation exchanges (d_0, d_1) with (d_1, d_0) and (a_0, a_1) with (a_1, a_0) simultaneously, so it does not disturb the comparison of the two family totals; the transpose does not affect either total. We therefore assume (10) from now on.

4. THE TWO BRANCH CERTIFICATES AND THE ESCAPE COMPARISON

4.1. What the certificates are. For a box $[l, u] \subseteq [0, 1]^8$, replace each of the 22 products $x_i x_j$ occurring in the right-hand sides of the moment equations by a fresh variable y_{ij} constrained by the four McCormick envelope inequalities

$$(11) \quad \begin{aligned} -y_{ij} + l_i x_j + l_j x_i &\leq l_i l_j, & -y_{ij} + u_i x_j + u_j x_i &\leq u_i u_j, \\ y_{ij} - u_i x_j - l_j x_i &\leq -u_i l_j, & y_{ij} - l_i x_j - u_j x_i &\leq -l_i u_j, \end{aligned}$$

which are exactly the expansions of $(x_i - l_i)(x_j - l_j) \geq 0$, $(u_i - x_i)(u_j - x_j) \geq 0$, $(u_i - x_i)(x_j - l_j) \geq 0$ and $(x_i - l_i)(u_j - x_j) \geq 0$. We also retain the redundant valid bounds

$$l_i l_j \leq y_{ij} \leq u_i u_j;$$

these are used when maximizing the partially dualized Lagrangian. Together with the budget row, the four chamber rows of (10), and the two rows $t \leq B/q^2$, $t \leq W/q^2$, this gives a linear programme in the $8 + 22 + 64 + 1 = 95$ variables (x, y, p, t) with 95 non-bound inequality rows and 42 equality rows, whose optimum dominates $\text{OPT}_{42}(x)$ for every x in the box.

A *branch certificate* is a rooted binary tree. A split node carries a coordinate index and an exact rational cut, and its two children are the parent box intersected with $x_d \leq c$ and with $x_d \geq c$ (the children overlap on the cutting hyperplane, which is harmless and gives an exhaustive cover). A leaf is one of three kinds:

- a *McCormick-dual leaf*, carrying an exact rational multiplier vector for the inequality rows (nonpositive) and an unrestricted one for the equality rows, together with the resulting exact rational bound u . The verifier reconstructs the partial Lagrangian, checks that the reduced cost of t vanishes and that the reduced costs of all 64 atom variables are ≤ 0 (so that the inner maximum over $p \geq 0$ is finite), evaluates the outer and product variables at their better box endpoints, recomputes the bound itself, and checks that it equals the stored u and satisfies the required threshold;
- an *exact empty leaf*, carrying one of five reasons (**rsort**, **csort**, **rcsum**, **dasum**, **budget**). Each is a purely interval-arithmetic obstruction to the chamber inequalities: e.g. **rsort** asserts $\max(l_{r_1}, l_{r_0}) > u_{r_1}$, which contradicts $r_0 \leq r_1$; and **budget** asserts that the sum of the four group minima forced by the preceding steps exceeds 4. These are exact rational emptiness tests, never solver infeasibility claims;
- a *plaid leaf*, asserting that the box lies wholly inside one of the four rational plaid neighbourhoods. With $s = \sqrt{3} - 1$, the four templates are obtained by choosing a square parity and a row parity; a box qualifies when, coordinate by coordinate, the template-0 coordinates have upper endpoint $\leq \frac{1}{10}$, the template-1 coordinates have lower endpoint $\geq \frac{19}{20}$, and the two template- s coordinates lie in $[\frac{3}{5}, \frac{7}{8}]$.

Lemma 18 (Branch-certificate inference). *Suppose every terminal box of a branch certificate is either empty, contained in a declared exceptional region, or carries a valid dual upper bound u . Then the root box is the union of the exceptional terminal boxes and boxes on which the relaxation has value at most the corresponding u .*

Proof. Induct upward from the leaves. At a split $x_d \leq c$ or $x_d \geq c$, the two closed child boxes cover the parent. The stated alternatives are therefore preserved at every internal node and hence at the root. $\square$

Definition 19. The canonical rational plaid neighbourhood is

$$(12) \quad N : \quad 0 \leq r_0, c_0, d_1, a_1 \leq \frac{1}{10}, \quad \frac{3}{5} \leq r_1, c_1 \leq \frac{7}{8}, \quad \frac{19}{20} \leq d_0, a_0 \leq 1$$

(in normalized coordinates). After the chamber restrictions (10), it is the only one of the four templates that survives.

4.2. The two certificates.

Theorem 20 (Global localization). *For every $x \in [0, 1]^8$ satisfying the chamber inequalities (10) and lying outside N ,*

$$\text{OPT}_{42}(x) \leq \frac{527}{1000}.$$

Theorem 21 (Local core). *For every $x \in N$ satisfying (10), either $\text{OPT}_{42}(x) \leq \frac{527}{1000}$, or the unnormalized counts satisfy the four aggregate core inequalities*

$$(13) \quad \frac{36}{25}q \leq r_1 + c_1 \leq \frac{37}{25}q, \quad r_0 + c_0 \leq \frac{9}{100}q, \quad d_1 + a_1 \leq \frac{9}{200}q, \quad (q - d_0) + (q - a_0) \leq \frac{1}{25}q.$$

Both theorems are established by exact branch certificates, distributed as the ancillary files `even_c527_sym_mc.json.gz` and `even_local_core_mc.json.gz`.

Remark 22 (Verification). Two programs sharing no code check these trees, both using only exact rational arithmetic and neither trusting any stored number it can recompute. The

delivered checker `verify_even_branch_certificate.py` rebuilds the moment system, every McCormick row, every dual, every split and every empty leaf, and ends in `ALL_EVEN_BRANCH_CERTIFICATES_OK`. The from-scratch verifier `verify_even_finite_independent.py` does not import it or the builders; it *discovers* the 42 rows by counting incidence fibres on real tori at $n = 8, 12, 16, 20$ (keeping a row only when all label tuples are attained with equal multiplicity, which is precisely the factorization statement, and finding by itself that $\{D, A\}$ is the unique ragged pair, thereby deriving the two aggregate rows rather than assuming them), re-validates them as exact identities on 200 random colourings at $n = 12, 16, 20, 28$ (8,400 equation checks), re-derives the McCormick rows, recomputes every leaf bound from its own partial Lagrangian, and re-proves every emptiness claim by its own interval reasoning. It agrees with the delivered checker on every count in Table 2, and rejects eight deliberately tampered copies of the first certificate. Appendix A gives commands and digests.

Proof of Theorems 20 and 21. The exact checks in Remark 22 establish that every internal node makes the declared exhaustive split and that every leaf has the stated type. In the first tree, the only exceptional leaves lie in N ; all other leaves are empty or have relaxation value at most $527/1000$. In the second, the only exceptional leaves satisfy (13); all other leaves are empty or have relaxation value at most $527/1000$. Applying Lemma 18 to the two root boxes proves the respective claims. $\square$

Their statistics are:

	Theorem 20	Theorem 21
root box	$[0, 1]^8$	N
nodes	6,929	4,423
split nodes	3,464	2,211
McCormick-dual leaves	3,368	1,639
plaid / core leaves	4	563
exact empty leaves	93	10
maximum depth	22	29
largest reconstructed leaf bound	$\frac{527}{1000} - 1.92 \cdot 10^{-6}$	$\frac{2329117}{4419584} = \frac{527}{1000} - 8.53 \cdot 10^{-7}$

TABLE 2. The two branch certificates. The 93 empty leaves of the first tree split as 1 `rsort`, 9 `csort`, 19 `rcsum`, 21 `dasum`, 43 `budget`; the 10 of the second are all `rcsum`. The four plaid leaves of the first tree all carry the template $(0, s, 0, s, 1, 0, 1, 0)$, i.e. the neighbourhood N ; the 563 core leaves of the second all lie wholly inside (13). The last row shows that both certificates are tight to about $2 \cdot 10^{-6}$.

4.3. The escape comparison.

Lemma 23. $\kappa - \frac{527}{1000} > \frac{1}{125}$ and $\gamma < \frac{1}{2}$. Consequently, for every $q \geq 64$,

$$(14) \quad H(2q) - \frac{527}{1000}q^2 > \frac{q^2}{125} - \frac{q}{2} - \frac{1}{4} > 0.$$

Proof. The inequality $\kappa - \frac{527}{1000} > \frac{1}{125}$ reads $4 - 2\sqrt{3} > \frac{535}{1000} = \frac{107}{200}$, i.e. $693 > 400\sqrt{3}$, i.e. $480249 > 480000$. The inequality $\gamma < \frac{1}{2}$ reads $\frac{1}{\sqrt{3}} > \frac{1}{2}$, i.e. $2 > \sqrt{3}$. The first inequality

of (14) is then Lemma 13 combined with these two comparisons. For the second, note that $q \geq 64$ gives

$$\frac{q^2}{125} - \frac{q}{2} - \frac{1}{4} \geq q\left(\frac{64}{125} - \frac{1}{2}\right) - \frac{1}{4} = \frac{3q}{250} - \frac{1}{4} \geq \frac{192}{250} - \frac{1}{4} > 0. \quad \square$$

Combining (9) with Theorems 20, 21 and Lemma 23:

Corollary 24. *Let $q \geq 64$ and let a configuration in the canonical chamber have $\min(B, W) \geq H(2q)$. Then its normalized profile lies in N and its unnormalized counts satisfy the aggregate core inequalities (13).*

5. THE LOCAL FINISH

Throughout this section $q \geq 130$, the configuration lies in the canonical chamber, its normalized profile lies in the neighbourhood N of (12), and the aggregate core (13) holds. We rename the eight counts to the local coordinates

(15) $u = r_0, \quad R = r_1, \quad v = c_0, \quad C = c_1, \quad e = d_1, \quad g = q - d_0, \quad f = a_1, \quad h = q - a_0,$
so u, v, e, f, g, h are small “defects” and R, C are close to sq . All eight are integers in $[0, q]$. Put

$$(16) \quad \begin{aligned} \sigma &= R + C, & m &= u + v, & a &= e + f, & b &= g + h, \\ \alpha &= 2q - \sigma - m, & \beta &= \sigma - q, & Q &= 2ef + 2gh, \\ X &= uv + RC, & Y &= (q-u)(q-C) + (q-R)(q-v). \end{aligned}$$

The core (13) reads

$$(17) \quad \frac{36}{25}q \leq \sigma \leq \frac{37}{25}q, \quad m \leq \frac{9}{100}q, \quad a \leq \frac{9}{200}q, \quad b \leq \frac{1}{25}q,$$

and the chamber (10) gives, in these coordinates,

$$(18) \quad u + R \leq v + C, \quad e + h \leq f + g, \quad \sigma + m + a - b \leq 2q.$$

Two remarks on notation. First, (X, Y) are exactly the two capacities of the H -profile $(r_0, r_1, c_0, c_1) = (u, R, C, v)$ in Definition 1: indeed $r_0c_1 + r_1c_0 = uv + RC = X$ and $(q - r_0)(q - c_0) + (q - r_1)(q - c_1) = (q - u)(q - C) + (q - R)(q - v) = Y$. Hence

$$(19) \quad \min(X, Y) \leq H(2q).$$

Second, α and β are the two “slacks” that the support cuts below will charge against.

5.1. The two support cuts. The proof uses exactly two members of an exactly certified library of 760 support-dual cuts for the moment system (Section 6 and Appendix A); they are entries 609 and 559 of that library, transported to the coordinates (15). A *support dual* is a pair $(\alpha_{\text{cut}}, \lambda)$ of exact rationals such that the corresponding combination of the 42 moment equations dominates $\alpha_{\text{cut}}B + (1 - \alpha_{\text{cut}})W$ columnwise on all 64 atoms; the resulting polynomial inequality then holds for every genuine configuration, with no relaxation and no branching. Cut 609 has $\alpha_{\text{cut}} = 1$ and cut 559 has $\alpha_{\text{cut}} = 0$, so they bound B and W separately:

Lemma 25 (Cuts 609 and 559). *For every configuration in the canonical chamber,*

$$(20) \quad B \leq F_B := X - \beta b + Q,$$

$$(21) \quad W \leq F_W := Y - \alpha a + Q.$$

In particular, writing $T := \min(F_B, F_W)$, we have $\min(B, W) \leq T$.

The two exact rational duals are shipped in `even_finite_support_cuts.json` and are byte-identical to entries 609 and 559 of the certified library `new_dual_cuts.json`; `verify_even_finite_algebra.py` recomputes both polynomial identities and the columnwise support property. Everything below is elementary algebra from (17), (18), (20) and (21).

5.2. Basic estimates and the defect dichotomy.

Lemma 26. *Under (17),*

$$(22) \quad \alpha \geq \frac{43}{100}q, \quad \beta \geq \frac{11}{25}q, \quad \alpha > a, \quad \beta > b.$$

Proof. $\alpha = 2q - \sigma - m \geq 2q - \frac{37}{25}q - \frac{9}{100}q = \frac{200-148-9}{100}q = \frac{43}{100}q$, and $\beta = \sigma - q \geq \frac{36}{25}q - q = \frac{11}{25}q$. Finally $a \leq \frac{9}{200}q < \frac{43}{100}q \leq \alpha$ and $b \leq \frac{1}{25}q < \frac{11}{25}q \leq \beta$. $\square$

Lemma 27 (Defect dichotomy). *Under (17),*

$$(23) \quad 2Q = 4ef + 4gh \leq a^2 + b^2 \leq \alpha a + \beta b, \quad \text{hence} \quad Q \leq \max(\alpha a, \beta b).$$

Proof. $4ef \leq (e+f)^2 = a^2$ and $4gh \leq (g+h)^2 = b^2$. By Lemma 26, $a^2 \leq \alpha a$ and $b^2 \leq \beta b$. Finally $Q \leq \frac{1}{2}(\alpha a + \beta b) \leq \max(\alpha a, \beta b)$. $\square$

This dichotomy is the only case distinction the local proof needs. We treat the two orderings of X and Y separately.

5.3. Case $X \leq Y$.

Proposition 28 (First defect case). *If $X \leq Y$ then $T \leq H(2q)$.*

Proof. If $Q \leq \beta b$ then $F_B \leq X$ by (20), so $T \leq X = \min(X, Y) \leq H(2q)$ by (19).

Assume therefore

$$(24) \quad Q > \beta b.$$

By Lemma 27, $Q \leq \alpha a$.

Step 1: a bound on F_B . Since $b \leq q/25$,

$$(25) \quad 2gh \leq \frac{b^2}{2} \leq \frac{qb}{50} \leq \beta b,$$

the last step by Lemma 26. Hence

$$(26) \quad F_B = X - \beta b + 2ef + 2gh \leq X + 2ef.$$

Step 2: four switched profiles. Consider the four H -profiles

$$(27) \quad (r_0, r_1, c_0, c_1) = (u+x, R, C, v+y), \quad (x, y) \in \{(e, 2f), (2e, f), (f, 2e), (2f, e)\}.$$

All coordinates lie in $[0, q]$: inside N we have $u, v \leq q/10$ and $e, f \leq q/10$, so the largest modified coordinate is at most $q/10 + 2q/10 = 3q/10$. The black capacity of (27) is

$$(28) \quad (u+x)(v+y) + RC = X + uy + vx + xy \geq X + xy = X + 2ef,$$

since $u, v, x, y \geq 0$ and each of the four choices in (27) has $xy = 2ef$. Combining with (26), each of the four profiles has black capacity at least F_B .

The white capacity of (27) is $(q-u-x)(q-C) + (q-R)(q-v-y) = Y - (x(q-C) + y(q-R))$, so the *white loss* is $x(q-C) + y(q-R)$. Put $L = 2q - \sigma$. Over the four choices in (27) the values of x sum to $3(e+f) = 3a$ and so do the values of y ; hence the average white loss is exactly

$$(29) \quad \frac{3a}{4}((q-C) + (q-R)) = \frac{3}{4}aL.$$

Step 3: the averaged loss is affordable. We claim

$$(30) \quad Q \leq a\left(\frac{L}{4} - m\right).$$

First, by (17),

$$(31) \quad \frac{L}{4} - m \geq \frac{1}{4}\left(2 - \frac{37}{25}\right)q - \frac{9}{100}q = \frac{13}{100}q - \frac{9}{100}q = \frac{q}{25}.$$

Second, from (24), Lemma 26 and (25),

$$2ef = Q - 2gh > \beta b - \frac{qb}{50} \geq \frac{11}{25}qb - \frac{qb}{50} = \frac{21}{50}qb.$$

Since $2ef \leq a^2/2$, this gives $b < \frac{25a^2}{21q}$, and therefore

$$(32) \quad Q \leq \frac{a^2 + b^2}{2} \leq \frac{a^2}{2} + \frac{qb}{50} < \frac{a^2}{2} + \frac{q}{50} \cdot \frac{25a^2}{21q} = \frac{11}{21}a^2 \leq \frac{11}{21} \cdot \frac{9q}{200}a = \frac{33}{1400}aq < \frac{1}{25}aq,$$

using $b^2/2 \leq qb/50$ (valid as $b \leq q/25$) and $a \leq 9q/200$. Comparing (31) and (32) proves (30).

Step 4: conclusion. Since $\alpha = L - m$, inequality (30) is equivalent to

$$(33) \quad \frac{3}{4}aL \leq a(L - m) - Q = \alpha a - Q.$$

By (29) at least one of the four profiles has white loss at most $\frac{3}{4}aL$, hence white capacity at least $Y - (\alpha a - Q) = F_W$. By (28) and (26) its black capacity is at least F_B . That profile therefore has H -value at least $\min(F_B, F_W) = T$, and so $T \leq H(2q)$. $\square$

5.4. Case $Y \leq X$.

Proposition 29 (Second defect case). *If $Y \leq X$ and $q \geq 130$, then $T \leq H(2q)$.*

Proof. If $Q \leq \alpha a$ then $F_W \leq Y$ by (21), so $T \leq Y = \min(X, Y) \leq H(2q)$ by (19).

Assume therefore

$$(34) \quad Q > \alpha a.$$

By Lemma 27, $Q \leq \beta b$.

Step 1: the integrality step. We claim $gh > 0$. If $gh = 0$, then (34) forces $a > 0$: otherwise Lemma 26 gives $e = f = 0$, and hence $Q = 0$, contradicting $Q > \alpha a = 0$. Using $a \leq \frac{9}{200}q$ and Lemma 26, we then have

$$Q = 2ef \leq \frac{a^2}{2} \leq \frac{9q}{400}a < \frac{43}{100}qa \leq \alpha a,$$

contradicting (34). Since g and h are integer counts, $g \geq 1$ and $h \geq 1$, so

$$(35) \quad b = g + h \geq 2.$$

This is the only point in the whole argument where integrality of the counts is used; it is also exactly the point at which the continuous relaxation is provably insufficient (Remark 30).

Step 2: averaging the two cuts. By (20) and (21),

$$(36) \quad T \leq \frac{F_B + F_W}{2} = \frac{X + Y}{2} - \frac{\alpha a + \beta b - 2Q}{2}.$$

By (23),

$$(37) \quad \alpha a + \beta b - 2Q \geq \alpha a + \beta b - (a^2 + b^2) = a(\alpha - a) + b(\beta - b) \geq b(\beta - b),$$

since $a(\alpha - a) \geq 0$ by Lemma 26. On the range $2 \leq b \leq q/25$ the function $b \mapsto b(\beta - b)$ is increasing, because its derivative $\beta - 2b \geq \frac{11}{25}q - \frac{2}{25}q = \frac{9}{25}q > 0$. Hence by (35),

$$(38) \quad \frac{\alpha a + \beta b - 2Q}{2} \geq \frac{2(\beta - 2)}{2} = \beta - 2.$$

Step 3: the balanced-plaid baseline. We claim

$$(39) \quad \frac{X + Y}{2} \leq \kappa q^2.$$

Normalize (u, R, C, v) by q and put, as in Lemma 12,

$$\rho = \frac{u + R}{q}, \quad \chi = \frac{v + C}{q}, \quad \xi = 1 - \rho, \quad \eta = 1 - \chi, \quad z = \left(\frac{u}{q} - \frac{R}{q}\right)\left(\frac{C}{q} - \frac{v}{q}\right).$$

The neighbourhood N gives $\rho \leq \frac{1}{10} + \frac{7}{8} < 1$ and $\chi < 1$, so $\xi, \eta > 0$. By (3),

$$\frac{X - Y}{q^2} = \rho + \chi - 2 - z, \quad |z| \leq \rho\chi, \quad \frac{X + Y}{q^2} = 1 + \xi\eta.$$

The assumption $X \geq Y$ forces $-\rho\chi \leq z \leq \rho + \chi - 2$, whence $2 - \rho - \chi \leq \rho\chi$, i.e. $2\xi + 2\eta - \xi\eta \leq 1$. With $w = \sqrt{\xi\eta}$, AM–GM gives $4w - w^2 \leq 1$, so $w \leq 2 - \sqrt{3}$ and

$$\frac{X + Y}{2q^2} = \frac{1 + w^2}{2} \leq \frac{1 + (2 - \sqrt{3})^2}{2} = 4 - 2\sqrt{3} = \kappa,$$

which is (39).

Step 4: conclusion. Combining (36), (38), (39) and $\beta \geq \frac{11}{25}q$,

$$(40) \quad T \leq \kappa q^2 - \frac{11}{25}q + 2.$$

By Lemma 13, $H(2q) \geq \kappa q^2 - \gamma q - \frac{1}{4}$, so (40) gives $T \leq H(2q)$ as soon as

$$(41) \quad \left(\frac{11}{25} - \gamma\right)q \geq \frac{9}{4}.$$

Now $\frac{11}{25} - \gamma = \frac{1}{\sqrt{3}} - \frac{14}{25}$, so (41) at $q = 130$ reads $\frac{130}{\sqrt{3}} - \frac{14 \cdot 130}{25} \geq \frac{9}{4}$, i.e. $\frac{130}{\sqrt{3}} \geq \frac{1501}{20}$, whose square is the integer comparison

$$6,760,000 \geq 6,759,003.$$

This holds, and the left-hand side of (41) is increasing in q (the coefficient $\frac{1}{\sqrt{3}} - \frac{14}{25} = 0.01735\dots$ is positive), so (41) holds for every $q \geq 130$. $\square$

Remark 30. The margin in (41) is $0.01735\dots$ per unit of q against a fixed deficit of $9/4$, which is why the threshold sits at $q = 130$ and not lower. The particular continuous relaxation used before the integrality step is not sharp at $n = 18$ and $n = 20$: it gives only $t \leq 43$ and $t \leq 53$, against the exact values 42 and 51. The integrality step (35) — and, in the finite range, the integer cut envelope of Section 6 — closes the remaining one- and two-unit gaps.

5.5. Proof of the upper bound for $q \geq 130$.

Theorem 31 (Large even orders). $t(2q) \leq H(2q)$ for every $q \geq 130$.

Proof. Suppose $t(2q) \geq H(2q) + 1$. By Lemma 6 there is a configuration with $\min(B, W) \geq H(2q) + 1$, and by Lemma 17 we may assume it lies in the canonical chamber. Since $q \geq 130 \geq 64$, Corollary 24 places its normalized profile in N and its counts in the aggregate core (13). Lemma 25 gives $\min(B, W) \leq T$, and Propositions 28 and 29 — which between them cover both orderings of X and Y — give $T \leq H(2q)$. Hence $\min(B, W) \leq H(2q)$, a contradiction. $\square$

Explicitly, a canonical profile falls into exactly one of three branches: outside N , where Theorem 20 and Lemma 23 give $\min(B, W) < H(2q)$; inside N but outside the aggregate core, where Theorem 21 and Lemma 23 give the same; and inside the core, where Section 5 gives $\min(B, W) \leq H(2q)$.

6. THE FINITE RANGE $q \leq 129$

6.1. The certified cut library. The support cuts used in Section 5 belong to a library of 760. A cut is a pair (θ, λ) with $\theta \in \mathbb{Q} \cap [0, 1]$ and $\lambda \in \mathbb{Q}^{42}$ such that, on each of the 64 atom columns,

$$(42) \quad \theta \mathbf{1}[\varepsilon = (1, 1, 1, 1)] + (1 - \theta) \mathbf{1}[\varepsilon = (0, 0, 0, 0)] \leq (\lambda^\top E)_{i,j,\varepsilon}$$

for every block (i, j) and atom $\varepsilon \in \{0, 1\}^4$, where E is the 42×64 atom-variable coefficient matrix of the moment equations. Its columns are indexed by (i, j, ε) .

Lemma 32 (Support-cut inference). *If (42) holds, then every genuine configuration satisfies*

$$\min(B, W) \leq \theta B + (1 - \theta)W \leq q^2 \lambda^\top b(x),$$

where $b(x)$ is the moment right-hand side at its outer profile.

Proof. The first inequality holds because the middle expression is a convex combination of B and W . Multiply the columnwise inequalities (42) by the nonnegative atom counts and sum. The moment equations replace Ep by $b(x)$. $\square$

The archive separates the library into 76 legacy polynomials in `benders_cuts.json` and 684 delivered rational duals in `new_dual_cuts.json`. An exact standard-library verifier reconstructs the 42 moment rows, checks the 64 column inequalities and polynomial identity for every delivered dual, and recovers rational witnesses for the 76 legacy cuts by exact simplex. It verifies all 760 cuts.

6.2. The cut envelope at a fixed order. For fixed q , evaluate each cut at integer outer counts, with constants scaled by q^2 and linear terms by q . Their minimum is the *cut envelope*. The checker searches boxes in $\{0, \dots, q\}^8$, propagates the chamber and budget inequalities, and bounds each multi-affine cut by exact McCormick bounds followed, when needed, by its exact corner maximum.

Lemma 33 (Integer-box inference). *Suppose such a search ends with no survivor and every nonempty terminal box has a cut bounded above by $H(2q) + \frac{1}{2}$ throughout the box. Then $t(2q) \leq H(2q)$.*

Proof. Every split partitions the integer points of its parent, so induction on the tree shows that every canonical integer profile belongs to a discharged terminal box. Lemma 32 gives $\min(B, W) \leq H(2q) + \frac{1}{2}$ there. Both $\min(B, W)$ and $H(2q)$ are integers, hence $\min(B, W) \leq H(2q)$. $\square$

Proposition 34. *For*

$$q \in \{3, 5\} \cup \{10, 11, \dots, 129\},$$

the cut envelope is at most $H(2q) + \frac{1}{2}$ throughout the canonical integer domain. Consequently $t(2q) = H(2q)$ for these orders.

Proof. We ran the released C++17 exact box engine on all 122 listed orders. Every search exhausted its stack with no survivor, visiting 13,379,092 nodes in total. The release verifier recompiles that engine, repeats every search, recomputes $H(2q)$ and compares every proof-relevant output field; thus it re-establishes the domain coverage required by Lemma 33.

The archived per-order JSON files bind the cut inputs and record witnesses, status fields and aggregate statistics, but are compact run summaries, not node-by-node tree certificates. Their checker audits those data and the full-tree accounting without claiming geometric coverage. As a separate implementation check, the Python engine produced the original summaries through $q = 60$ and was rerun for every $61 \leq q \leq 129$; its non-runtime envelope outputs agree exactly with the C++ outputs. The two engines implement the same box algorithm, so this agreement is corroborative rather than a conceptually independent proof method. Lemma 33 gives the upper bound, and Proposition 10 gives equality. $\square$

The orders $q = 1, 2$ are elementary. The four remaining envelope exceptions $q = 4, 6, 7, 9$ are direct finite searches, while $q = 8$ is covered by the union-domain verification of $n = 16$. They are proved together with the odd searches in Section 7.

7. THE EXHAUSTIVE FINITE SEARCHES

On an odd torus the map φ of Lemma 7 is bijective, so label parity carries no geometric information. We now give the finite arguments for $t(15)$ and $t(17)$ and for the five even orders not covered by Proposition 34.

7.1. The common search reduction. For odd n , write

$$(s_0, s_1, s_2, s_3) = (|R|, |C|, |D|, |A|), \quad S = \sum_i s_i.$$

If a target M is feasible, Lemmas 8, 7 and 9 imply

$$(43) \quad S \leq 2n,$$

$$(44) \quad s_i s_j \geq M, \quad (n - s_i)(n - s_j) \geq M \quad (i < j),$$

$$(45) \quad n^2 - n(x + y + z) + xy + xz + yz \geq 2M \quad \text{for each triple } x, y, z.$$

These necessary conditions give a finite profile list. We quotient only by explicit affine symmetries: transpose, $(r, c) \mapsto (r, -c)$, and, for odd n , $(r, c) \mapsto (r + c, r - c)$. No classification of the full affine symmetry group is used.

For each surviving profile, fix two line families, say R, C , up to a subgroup of their affine stabilizer. For an enumerated set A , define

$$\begin{aligned} p_d &= \#\{(r, c) : r \in R, c \in C, r - c = d, r + c \in A\}, \\ q_d &= \#\{(r, c) : r \notin R, c \notin C, r - c = d, r + c \notin A\}, \quad Q_0 = \sum_d q_d. \end{aligned}$$

For any D of its prescribed size,

$$(46) \quad |B| = \sum_{d \in D} p_d, \quad |W| = Q_0 - \sum_{d \in D} q_d.$$

Thus the last family is an exact fixed-cardinality, two-constraint subset problem.

Lemma 35 (Exhaustive-search inference). *Assume that the profile filter uses only necessary conditions, the fixed-pair list meets every orbit of a subgroup preserving the searched profile domain, every admissible third-family set is enumerated, and (46) is decided exactly for the last family. If every listed profile is infeasible, then the target M is infeasible.*

Proof. Any feasible line colouring has a profile in the filtered list. A permitted affine symmetry sends its fixed pair to an enumerated representative and transports the remaining line sets within the searched domain. Its third set occurs in the enumeration, and its last set satisfies (46); an exact decision procedure must therefore find it. $\square$

7.2. The value $t(15)$. The line sets

$$\begin{aligned} R &= \{5, 7, 8, 11, 12, 14\}, & C &= \{0, 3, 4, 7, 9, 10, 12, 13\}, \\ D &= \{0, 1, 4, 8, 12, 13, 14\}, & A &= \{1, 2, 3, 9, 10, 12, 14\} \end{aligned}$$

have $|B| = 20$ and $|W| = 22$, proving the lower bound.

At target $M = 21$, conditions (43)–(45) leave 247 canonical profiles. Their counts in the strata $S = 20, \dots, 30$ are

$$1, 1, 4, 6, 13, 18, 29, 37, 47, 52, 39.$$

For each profile, the fixed-pair subgroup is generated by independent translations, common multiplication by a unit, transpose when the fixed sizes agree, and $(R, C) \mapsto (R, -C)$ only when the two completion sizes agree. The last restriction ensures that the reflection does not exchange two distinct ordered profiles. A separate audit regenerates a canonical identifier for every orbit and checks all 43 fixed-pair orbit types by Burnside’s lemma.

The first enumerator decides (46) by depth-first search, using necessary cardinality bounds on each suffix. An independent enumerator chooses a different fixed pair on 19 profiles and uses an exact $7 + 8$ meet-in-the-middle Pareto table. Their respective case totals are

$$6,074,753,568 \quad \text{and} \quad 6,241,793,402.$$

Both return UNSAT on all 247 profiles; the complete second result file is checked for exact profile coverage and unique UNSAT records.

Theorem 36. $t(15) = 20$.

Proof. The displayed line sets give $t(15) \geq 20$. The profile generation, orbit audit, complete third-family enumeration and the two exact last-family decisions satisfy Lemma 35 at target 21. Hence $t(15) \leq 20$. $\square$

7.3. A union-domain verification at $n = 16$. Although Theorem 2 already gives $t(16) = 32$, a direct search is useful both as an independent small-order proof and as the integration of the earlier $n = 16$ computation. Let $E, O \subset \mathbb{Z}/16\mathbb{Z}$ be the even and odd residue classes and use profiles

$$(r, c, d_0, d_1, a_0, a_1) = (|R|, |C|, |D \cap E|, |D \cap O|, |A \cap E|, |A \cap O|).$$

The budget, pair bounds and triple identity leave exactly 1,898 ordered profiles at target 33.

The proof search groups these profiles only by (r, c) , after transposing so that $r \leq c$, and searches the *union* of all associated completion profiles. This union is invariant under simultaneous parity exchange and under exchanging D with A . Hence independent translations, common multiplication by an odd unit, $(R, C) \mapsto (R, -C)$, and transpose when $r = c$ act within the searched domain; no parity-orientation transfer is needed. Fixed-pair representatives are selected by a lexicographic canonical-form test on translation necklaces.

The enumerator scans every admissible A -mask. Only the two necessary row–column–antidiagonal triple inequalities are used as a scalar prefilter; for every survivor it scans every admissible D -mask literally. The exact totals are

fixed-pair representatives	159,551
A -masks tested	3,226,530,570
A -masks surviving the scalar filter	2,951
diagonal-cardinality cases	29,768
D -masks tested	38,830,322.

None is feasible. A separately written Python audit regenerates the profile domain, proves its closure under the three domain symmetries, computes the seven pair-orbit counts by Burnside’s lemma, regenerates every representative list and hash, and checks the certificate totals. It prints `UNION_AUDIT_OK`. The older 677-job programs, which share their search chassis but use different completion kernels, reproduce the same upper bound as a secondary cross-check.

Proposition 37. *The union-domain search certifies that no $33 + 33$ placement exists on the 16×16 torus.*

Proof. The 1,898 profiles contain every feasible colouring by the necessary conditions. The union domain makes every quotient symmetry profile preserving, and the audited representative list meets every fixed-pair orbit. All third-family masks and then all admissible last-family masks are tested, so Lemma 35 applies. $\square$

7.4. The remaining orders.

Lemma 38 (Parity-refined even sweeps). *Let $n = 2h$, and suppose a line colouring has target M . With $P_\varepsilon = \{x \in \mathbb{Z}/n\mathbb{Z} : x \equiv \varepsilon \pmod{2}\}$, give it the profile*

$$(r, c, d_0, d_1, a_0, a_1) = (|R|, |C|, |D \cap P_0|, |D \cap P_1|, |A \cap P_0|, |A \cap P_1|).$$

Put $s_R = r$, $s_C = c$, $s_D = d_0 + d_1$, $s_A = a_0 + a_1$. For distinct families X, Y , define

$$J_{DA} = 2(d_0 a_0 + d_1 a_1),$$

$$\bar{J}_{DA} = 2((h - d_0)(h - a_0) + (h - d_1)(h - a_1)),$$

$$J_{XY} = s_X s_Y, \quad \bar{J}_{XY} = (n - s_X)(n - s_Y) \quad (\{X, Y\} \neq \{D, A\}).$$

After possibly complementing all four line sets, every feasible profile satisfies

$$(47) \quad \sum_X s_X \leq 2n, \quad J_{XY}, \bar{J}_{XY} \geq M, \quad n^2 - n(s_X + s_Y + s_Z) + J_{XY} + J_{XZ} + J_{YZ} \geq 2M$$

for every pair and every triple of distinct families.

It is enough to retain one profile from each orbit generated by exchanging R, C , exchanging D, A , simultaneously swapping d_0, d_1 and a_0, a_1 , and, on the boundary $\sum_X s_X = 2n$, complementation. For each retained profile, choose an R, C and D, A orientation and search both simultaneous parity orientations unless they coincide. Fixing R, C , one may quotient by independent translations and common multiplication by a unit, as well as transpose when $r = c$ and $(R, C) \mapsto (R, -C)$ when $(d_0, d_1) = (a_0, a_1)$. Exhausting the resulting pair representatives, all A with the prescribed parity sizes, and the exact parity-constrained D -completion problem exhausts the target M .

Proof. The first inequality in (47) is Lemma 8. Lemma 7 gives J_{XY} and $\bar{J}_{XY}$ as the black and white pair-intersection counts; the exceptional formulas for D, A use its two-or-zero fibres. Since B and W lie in the corresponding intersections, the pair inequalities follow. Applying Lemma 9 to each triple gives the last inequality.

Transpose, $(r, c) \mapsto (r, -c)$, and translation by one row induce the three profile exchanges above; complementation swaps the armies. An independent translation of R, C shifts the D, A labels by amounts of the same parity, so it either preserves both parity orientations or exchanges both. Every unit modulo $2h$ is odd and preserves parity. Transpose and the relative-sign map preserve a job under precisely the stated size conditions. Thus the union of the paired orientations is stable under the fixed-pair quotient and meets every feasible pair orbit. For each representative, the third-family enumeration uses only necessary suffix bounds. Every surviving A is passed to the exact subset problem (46), now with the two constraints $|D \cap P_\varepsilon| = d_\varepsilon$. Lemma 35 then gives the asserted coverage. $\square$

The same profile-orbit-completion programme was implemented for arbitrary $n \leq 18$, using Lemma 38 at even orders. Its independent audit regenerates every worklist, the Burnside orbit counts and the complete candidate-space arithmetic. The upper campaigns needed here are:

n	target	jobs	third-family cases
8	9	6	720
12	19	100	5,111,848
14	26	76	70,259,455
17	29	316	157,158,165,660
18	43	259	228,644,005,008

Every job is UNSAT. For $n = 17$, a second implementation with a different quotient and enumeration order also returns UNSAT on all 316 jobs, after 158,802,287,408 logical third-family cases.

At $n = 17$ the line sets

$$\begin{aligned} R &= \{0, 1, 2, 4, 5, 7, 8, 9\}, & C &= \{0, 1, 2, 4, 5, 7, 8, 9, 13\}, \\ D &= \{2, 3, 4, 7, 10, 13, 14, 15\}, & A &= \{2, 5, 6, 7, 9, 11, 12, 13, 16\} \end{aligned}$$

have $|B| = |W| = 28$.

Theorem 39. $t(17) = 28$.

Proof. The displayed sets give the lower bound. The audited 316-job campaign meets the hypotheses of Lemma 35 at target 29, giving the upper bound. $\square$

Proposition 40. For $q \in \{1, 2, 4, 6, 7, 8, 9\}$, one has $t(2q) = H(2q)$.

Proof. The cases $q = 1, 2$ follow by inspection. For $q = 4, 6, 7, 9$, the first, second, third and fifth rows of the campaign table exclude $H(2q) + 1$ queens of each colour. Proposition 37 does the same for $q = 8$. Proposition 10 supplies every lower bound. $\square$

Proof of Theorems 2 and 5. Proposition 10 gives $t(2q) \geq H(2q)$. The reverse inequality is Theorem 31 for $q \geq 130$, Proposition 34 for $q \in \{3, 5\} \cup \{10, \dots, 129\}$, and Proposition 40 for the remaining q . Theorems 36 and 39 give Theorem 5. $\square$

Proof of the corollaries. Corollary 3 follows from Theorem 2 and Lemmas 12 and 13. The displayed values follow by evaluating the finite maximum (1). $\square$

The odd construction below explains some of the sensitivity to divisors, but does not suggest an odd analogue of Theorem 2.

Proposition 41 (Divisor lifting). *If $d \mid n$ then $t(n) \geq (n/d)^2 t(d)$.*

Proof. Let (R, C, D, A) be an optimal configuration on the $d \times d$ torus and let $\pi : \mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/d\mathbb{Z}$ be reduction. Take $(\pi^{-1}R, \pi^{-1}C, \pi^{-1}D, \pi^{-1}A)$. A cell (r, c) of the n -torus is black-homogeneous iff $(\pi r, \pi c)$ is, since $\pi(r - c) = \pi r - \pi c$ and $\pi(r + c) = \pi r + \pi c$; and each cell of the d -torus has exactly $(n/d)^2$ preimages. So both armies scale by $(n/d)^2$, and Lemma 6 applies. $\square$

For example, $t(39) \geq 144$, $t(45) \geq 180$ and $t(51) \geq 252$. The lift from proper divisors is not generally sharp:

$$\max_{\substack{d \mid 9 \\ d < 9}} (9/d)^2 t(d) = 0 < 7 = t(9), \quad \max_{\substack{d \mid 15 \\ d < 15}} (15/d)^2 t(d) = 18 < 20 = t(15).$$

The odd asymptotics remain open; the upper bound $t(n) \leq n^2/8$ is [2, Thm. 1.4].

8. USE OF AI TOOLS

Large-language-model tools assisted with conjecture generation, drafting, code generation and source review. The author checked the final mathematical arguments and proof-critical outputs and takes responsibility for the paper. Agreement among AI-assisted implementations is treated as an error-detection measure, not as external scholarly reproduction.

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APPENDIX A. VERIFICATION AND REPRODUCTION

The complete ancillary archive is bound by SHA256SUMS. An immutable public snapshot is available at

<https://github.com/jphme/math-problems/tree/459046273420a61d51bac18f3a260b4317ec2e62/peaceable-queens-even-torus>.

The following command, run from `anc/`, replays the exact proof objects, reruns the complete finite cut envelope and audits the frozen outputs of the larger searches:

```
uv run --isolated --python 3.14.6 \
  --with sympy==1.14.0 --with mpmath==1.3.0 \
  python verify_bundle.py
```

It ends with

PEACEABLE_QUEENS_RELEASE_BUNDLE_OK.

The recorded environment is CPython 3.14.6 and SymPy 1.14.0; all other Python checks use only the standard library. The sources use C++17 or C++20 as specified by their build commands and were rebuilt with Apple Clang 21.0.0; the union-domain enumerator was also reproduced with GCC 14 and Clang 17. No floating-point comparison affects a proof verdict.

A.1 Exact proof objects. The files `even_c527_sym_mc.json.gz` and `even_local_core_mc.json.gz` contain the two branch trees. The delivered checker and a separately written verifier both reconstruct every split, moment row, McCormick inequality and rational leaf dual. The latter also derives the 42 moment rows from torus incidence fibres and tests them on explicit colourings.

The files `new_dual_cuts.json` and `benders_cuts.json` together contain the 760 support cuts. `verify_support_duals.py` verifies all of them exactly and recovers the 76 legacy duals independently. The two cuts used in Section 5 are also checked by `verify_even_finite_algebra.py`; the independent prose checker verifies the displayed algebra and exact constants.

A.2 Finite computations. The directories `records_q003_q020/` and `records_q021_q129/` contain compact summaries of the 122 ladder runs. They do not contain the split nodes or terminal boxes. `check_envelope_records.py` recomputes H and checks the order set, schemas, source and input hashes, thresholds, witnesses and aggregate node/leaf accounting. Full coverage is re-established by

`python replay_finite_envelope.py`

This compiles `box_engine.cpp` with strict C++17 flags and repeats all 122 searches. The separately written Python implementation was also rerun for $q = 61, \dots, 129$; its exact agreement with the C++ output is frozen in `finite_envelope_python_crosscheck_q061_q129.json`. The ledger checker audits that frozen cross-check, while `crosscheck_finite_backends.py` recreates it. These programs share the same exact integer-box algorithm.

Range	Full computation	Action in the one-command verification
$q = 3, 5, 10, \dots, 129$	<code>box_engine.cpp</code>	full exact rerun and comparison with every frozen summary
$n = 15$	<code>peace15_solver.cpp</code> , <code>profile_enum.cpp</code>	profile, orbit, output-coverage and witness audits
$n = 16$	<code>peace16_union_enum.cpp</code>	domain, representative, hash, total and witness audits
$n = 8, 12, 14, 17, 18$	<code>general_sweep/solver_b.cpp</code>	worklist, orbit, output-coverage and witness audits

TABLE 3. Verification modes for the finite searches. The README gives the full rebuild commands for the computations audited only through frozen outputs here.

For $n = 15$, the archive includes both enumerator sources, both complete outputs, the 247-profile worklist, a Burnside audit and direct witness checks. For $n = 16$, it includes the union-domain source, certificate and independent representative-list audit, as well as the older two-kernel cross-check. The directory `general_sweep/` contains the general- n source, audits and frozen outputs used for $n = 8, 12, 14, 17, 18$, together with the second $n = 17$ implementation and output.

The rational branch trees and support duals are standalone proof objects. The ladder summaries and exhaustive-search outputs are audit records: their logical force is the corresponding short source together with the reductions proved in Sections 6 and 7. The release command reruns the complete ladder; the larger board enumerations are reproducible with the commands in `README.txt`.

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